Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 03
Chapter 1: Real Numbers

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. If p is a prime number, then the LCM of p and p^2 is: [1]
(a) p (b) p^2 (c) p^3 (d) 1

Q2. If $\text{HCF}(a, b) = 12$ and $a \times b = 1800$, then $\text{LCM}(a, b)$ is: [1]
(a) 3600 (b) 150 (c) 900 (d) 600

Q3. If n is any natural number, then $6^n - 5^n$ always ends with the digit: [1]
(a) 1 (b) 3 (c) 5 (d) 7

Q4. If n is a natural number, then $(9^{2n} - 4^{2n})$ is always divisible by: [1]
(a) 5 only (b) 13 only (c) both 5 and 13 (d) neither 5 nor 13

Q5. **Assertion (A):** The number 6^n cannot end with the digit 0 for any natural number $n \in \mathbb{N}$.
Reason (R): The prime factorisation of 6 has only 2 and 3 as prime factors, and does not contain the prime factor 5. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Check whether 4^n can end with the digit 0 for any natural number $n \in \mathbb{N}$. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Prove that $\sqrt{3}+\sqrt{5}$ is an irrational number. [3]

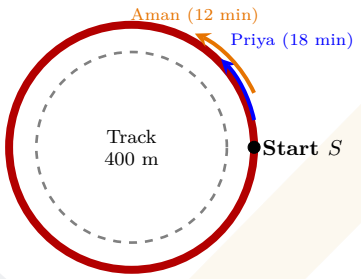
Q8. Prove that $\frac{2+3\sqrt{5}}{7}$ is an irrational number, given that $\sqrt{5}$ is irrational. [3]

Q9. Prove that $3+\sqrt{2}$ is irrational. [3]

SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

Q10. CASE STUDY: CASE STUDY 03: PRECISION ATHLETICS CIRCULAR RUNNING TRACK [4 Marks]

At an inter-state sports complex, an Olympic standard circular synthetic track has a perimeter of 400 m. Two athletes, Priya and Aman, start simultaneously from the same starting marker S and run in the same direction at uniform speeds. Priya takes 18 minutes to complete one full lap, while Aman takes 12 minutes to complete one full lap.



(i) After how many minutes will Priya and Aman meet again at the starting point S? [2]

(ii) How many complete laps will Priya and Aman have run respectively by the time they meet at S? [1]

(iii) If a third athlete Rohan joins them taking 24 minutes per lap, after how many minutes will all three meet at S? [1]

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